

SESA 6067

Spacecraft Orbital Mechanics

Author: Yusaf Sultan

Lecturer: Alexander Wittig

Semester: Semester 2 (Jan 26 - Mar 26)

Word Count: 580

Contents

1. Chapter 1: Orbital Manuevers	3
1.1. Manuevers	3
1.2. Types of Manuevers	3
1.3. Impulsive Manuever Assumption	3
1.4. Hohmann Transfers	3
1.4.1. Circular Hohmann Transfer	3
1.4.2. Elliptical Hohmann Transfer	5

List of Weeks

List of Figures

Figure 1 Schematic diagram of a circular Hohmann Transfer.	4
Figure 2 Schematic diagram of elliptical Hohmann transfers.	5

List of Tables

1. Chapter 1: Orbital Maneuvers

1.1. Maneuvers

A maneuver is defined as **A controlled change in the velocity (magnitude and/or direction) of a spacecraft to alter its orbit**. Each mission contains multiple maneuvers and those maneuvers can be made of the following building blocks:

- Change orbit shape.
- Change orbit orientation (plane).
- Change of orbit phasing (anomaly).

There can be multiple reasons for maneuvers within a mission, including orbital transfers, maintenance and disposal. The goal of mission design is to **minimize ΔV within mission constraints**, these constraints being:

- Transfer time.
- Forbidden regions (e.g. radiation belts).
- Pointing requirements (panels, instruments).

1.2. Types of Maneuvers

This module will cover various types of coplanar transfers and plane change maneuvers, these are:

- Coplanar transfers
 - (Bi-elliptic) Hohmann transfer
 - Phasing maneuvers
 - Non-Hohmann transfers
 - Apse line rotation
- Plane change maneuvers
 - Inclination change
 - RAAN change

1.3. Impulsive Maneuver Assumption

For this module, each maneuver is assumed to be impulsive (kick maneuver). This assumes that the burn time is negligible in comparison to the orbital time. It is important to note that **ΔV cost of a mission is the sum of all maneuvers**. Further note that for this topic **ΔV is a vector and so must be added correctly**.

1.4. Hohmann Transfers

A Hohmann transfer is a form of co-planar maneuver that can be used to raise or lower an orbit. These types of maneuvers are typically used during launch to increase the height of an orbit to the desired altitude.

1.4.1. Circular Hohmann Transfer

A circular Hohmann transfer is used to move between two circular orbits via one intermediate elliptical orbit. The derivation for the orbital velocities of each orbit can be done via the angular momenta of each orbit. To start with a diagram of a circular Hohmann transfer is shown in **Figure 1**.

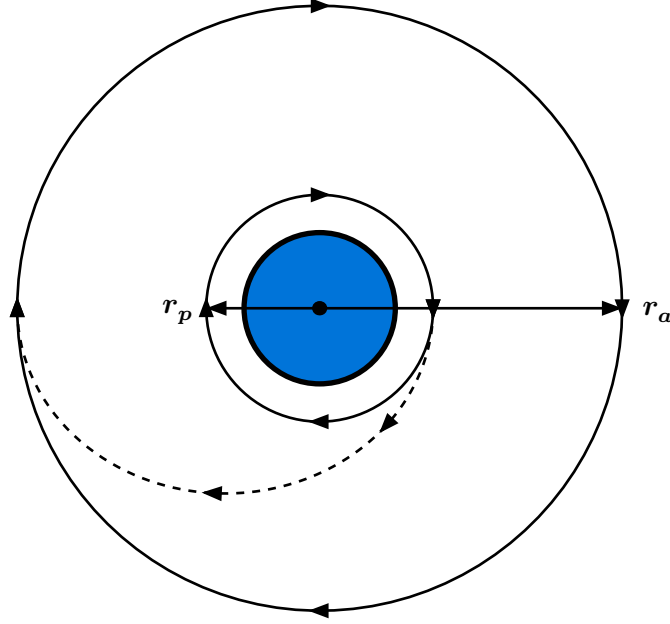


Figure 1: Schematic diagram of a circular Hohmann Transfer.

In order to calculate the ΔV required for the two burns in this maneuver the polar orbit equation can be used, shown in **Eq. 1**, note that at perigee for the elliptical transfer orbit $\theta = 0^\circ$

$$r = \frac{p}{1 + e \cos(\theta)} = \frac{a(1 - e^2)}{1 + e \cos(\theta)} = \frac{h^2/\mu}{1 + e \cos(\theta)} \rightarrow r_p = \frac{h^2/\mu}{1 + e} \quad (1)$$

The expression in **Eq. 1** can be combined with the eccentricity equation shown in **Eq. 2**.

$$e = \frac{r_a - r_p}{r_a + r_p} \quad (2)$$

Combining both expressions **Eq. 1** and **Eq. 2** yield the following expression shown in **Eq. 3**.

$$r_p = \frac{a(1 - e^2)}{1 + e} \quad e = \frac{r_a - r_p}{r_a + r_p} \rightarrow h = \sqrt{2\mu \frac{r_a r_p}{r_a + r_p}} \quad (3)$$

Where h is the specific angular momentum ($m^2 s^{-1}$) of the orbit and r_p and r_a are the radius at perigee and apogee shown in **Figure 1**. Now consider the general expression for specific angular momentum shown in **Eq. 4**.

$$h = \vec{r} \times \vec{v} = rv \sin(\phi) \quad (4)$$

Where $\vec{r}$ and $\vec{v}$ are the position and velocity vectors and the angle ϕ is the angle between these two vectors. Importantly, $\phi = 0^\circ$ **at the apogee and perigee points in the elliptical transfer orbit**. Combining **Eq. 3** and **Eq. 4** the velocities at r_p and r_a can be written in the form shown in **Eq. 5**.

$$v_a = \frac{h}{r_a} \quad v_p = \frac{h}{r_p} \quad \text{Where } h = \sqrt{2\mu \frac{r_a r_p}{r_a + r_p}} \quad (5)$$

Note that if $r_p = r_a$, meaning the transfer orbit is circular, then the expression simplifies to the following known expressions for circular orbits, shown in **Eq. 6**.

$$\text{If } r_a = r_p = r \quad \text{Then} \quad h = \sqrt{\mu r} \quad v = \sqrt{\frac{\mu}{r}} \quad (6)$$

1.4.2. Elliptical Hohmann Transfer

The derivation on the previous page is **not specific to circular orbits** as all of the equations used are general orbital equations. For this reason the equations can also be used for **transferring between two elliptical orbits** given they are coaxial (the Apse line of both orbits is the same).

However, there is a caveat in that it is not initially clear which transfer orbit is the most efficient. This is exacerbated if the two orbits are not in the same direction. This scenario would require calculating the deltaV of all 4 possible orbits (2 transfer orbits in each direction). The two possible orbits are shown in **Figure 2**.

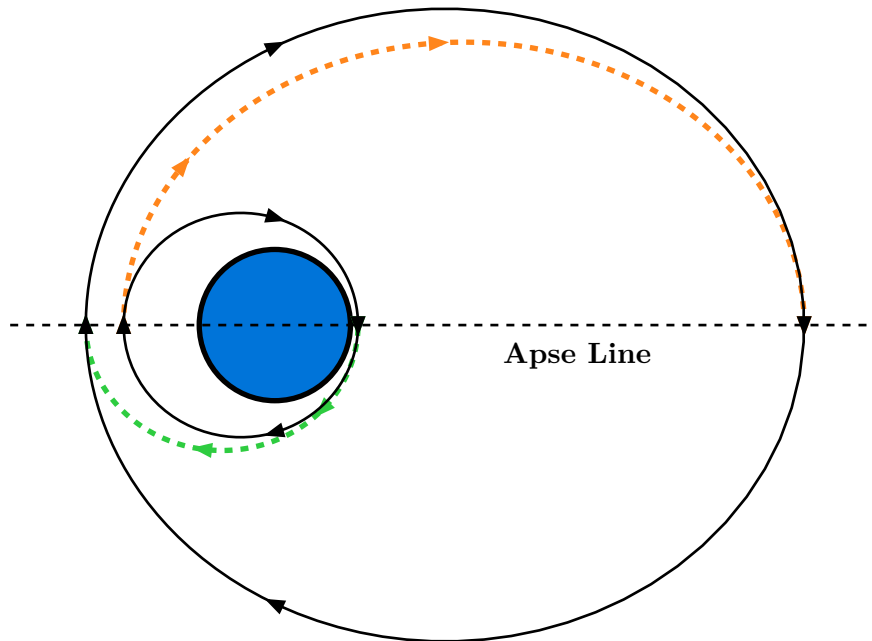


Figure 2: Schematic diagram of elliptical Hohmann transfers.

